

# Project: SQL Exploratory Data Analysis - Summary of Findings

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## 1. Project Overview

This project analyzes the AdventureWorksDW2025 dataset using SQL to explore sales performance, customer behavior, and product performance. The analysis combines data from multiple fact and dimension tables to generate business insights. The scope of our project includes

- Understand database structure and relationships
- Analyze customer demographics and behavior
- Evaluate product and category performance in terms of sales
- Measure key business KPIs
- Identify sales trends over time (YoY, MoM)
- Analyze geographical performance in terms of sales or revenue

## 2. Database & Data Structure Insights

- There are in total 31 Tables and 5 Views in the DataWarehouse. However, not all tables and views are relevant for our analysis. We have selected the tables which are relevant for our analysis
- Relevant Fact tables: FactInternetSales, FactResellerSales (Combined with View)
- Relevant Dimension tables: Customer, Product, Product Category, Product SubCategory, Geography, Sales Territory
- Relationships between tables are well-defined through primary and foreign keys.
- No major structural inconsistencies were identified.
- The database follows a star schema with clearly defined Fact and Dimension tables.

## 3. Data Quality & Validation

- No null values found in critical date fields (OrderDate, ShipDate, DueDate).
- Date relationships are logically consistent ( $\text{OrderDate} \leq \text{ShipDate} \leq \text{DueDate}$ ).
- Data is reliable for further analysis.
- Abbreviations exist. However, replaced with full form while analyzing.

## 4. Customer Insights

- Customers are distributed across multiple countries and cities. Countries are United States, United Kingdom, Germany, France, Australia, and Canada
- Gender distribution: Male – 9351 (around 51%), Female – 9133 (around 49%)
- Age distribution: Majority of customers fall into Age Group 46-60 (around 48%) followed by Age Group 60+ (around 35%) and Age Group 31-45 (around 16%). We can see that all of our customers are either old age or middle age. We don't have

any young customers (age under 30) yet. Our youngest customer is 40 years old and oldest customer is 110 years old which also advocates the age group distribution truth.

- Key demographics include education level and occupation categories. Our customers have varied education levels from high school to graduate level. Customer occupation includes Professional, Skilled Manual, Management, Clerical, and Manual
- Customer tenure (Average 13 Years) and age at first purchase (Average Age at first Purchase 43 Years) indicate long-term engagement patterns.

## **5. Product Insights**

- Products are structured into categories and subcategories. We have total 4 categories, 37 Sub-categories, and 606 Products. 4 Categories are Bike, Component, Clothing, and Accessories.
- Product attributes such as size, color, and product line provide segmentation opportunities.
- Top-performing products: Mountain-200 Black, 38; Mountain-200 Black, 42; Mountain-200 Silver, 38; Mountain-200 Silver, 42; Mountain-200 Silver, 46
- Lowest-performing products: LL Road Seat/Saddle; Mountain Bike Socks, L; LL Touring Frame - Blue, 58; LL Mountain Frame - Black, 40; LL Touring Seat/Saddle

## **6. Geographic Insights**

- Top countries by customer count: United States> Australia> United Kingdom > France> Germany> Canada
- Top countries by sales: United States> Canada> Australia> United Kingdom > France> Germany
- Top 10 cities generating highest revenue: London, Burnaby, Paris, Berlin, Newton, Montreal, Toronto, Ottawa, Shawnee, Mississauga
- Bottom 10 cities generating lowest revenue: Braunschweig, Eilenburg, Kassel, Kiel, Saarbrücken, Mühlheim, Muehlheim, Münster, Hannover, Offenbach

\* Note: Even though Canada has lowest number of customers, it takes second position in terms of sales or revenue. That means customers from Canada either order more quantity or order high-value products.

## **7. Operational Insights**

- Average shipping time: 7 days
- Delivery gap between shipping and due date: 5 days
- Indicates efficiency of logistics and fulfillment processes.

## 8. Time-Based Trends

- Sales trends analyzed across years and months.
- Year-over-Year (YoY) Trend: Sales demonstrated consistent year-over-year growth across the full date range, with acceleration in later periods. Total Sales increased by 34.69% in 2012 compared to 2011 and 46.68% in 2013 compared to 2012. Year 2010 and 2014 excluded from the comparison as they don't contain whole year data.
- Month-over-Month (MoM) Trend: There is no strong signal that shows our revenue depends on seasonality, rather our analysis finds that Month-over-Month Sales fluctuate a lot.

## 9. Sales & Performance Metrics (KPI)

| Metric               | Value              |
|----------------------|--------------------|
| Total Customers      | 18,484             |
| Total No. of Orders  | 1,21,253           |
| Total Order Quantity | 2,74,776           |
| Total Sales          | \$ 1,09,809,274.20 |
| Avg Product Price    | \$ 802,11          |

## 10. Key Business Insights [Summary]

- Sales demonstrated consistent year-over-year growth across the full date range, with acceleration in later periods. Total Sales increased by 34.69% in 2012 compared to 2011 and 46.68% in 2013 compared to 2012
- There is no strong signal that shows our revenue depends on seasonality, rather our analysis finds that Month-over-Month Sales fluctuate a lot
- Product Categories "Bikes" and "Components" generate nearly 96% of our revenue. The contribution from products falling into "Clothing" and "Accessories" is insignificant, around 4% combined.
- All our current customers are either old age or middle age. We don't have any young customers (Age under 30) yet. Opportunities exist to reach out to this specific group.
- Our customers come from only six countries. Huge opportunities exist to expand the business across other countries or continents.
- Overall, business is doing good in terms of sales or revenue, however opportunities exist to optimize underperforming products as well as expand customer base and regions.